

Carson Slater

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SUMMARY

Motivated graduate student seeking consulting opportunities. Aspiring applied statistician who loves to use statistical methods to empower people with insights for good stewardship.

PAPERS

Assessing the Effects of Surface-Stabilized Zero-Valent Iron Nanoparticles on Diverse Bacteria Species Using Complementary Statistical Models (March 2025) [Paper](#)

- Used ordinal regression and generalized linear models to understand relationships between nanoparticle coatings and bacterial inhibition.

Assessing Interference with Regression Analysis Techniques (April 2024) [Paper](#)

- Used generalized additive models to predict pairwise interference in a Dynamic Spectrum Access context.
- Paper was presented at the [2024 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems \(WMCS\)](#).

PROJECTS

Mo(Wa)²TER Introduction to Data Science Course Material [Harvard Dataverse](#)

- Edited and compiled material with Dr. Mandy Hering on her contributions to the National Science Foundation Harnessing the Data Revolution: Data Science Corps project (Grant no. 1924146).

Forecasting Demand With Generalized Additive Models [Link to Github](#)

- Built time-series forecasting models to predict quantity-demanded for fast-moving consumer goods in Indonesia.
- Presented poster at University of Illinois – Chicago’s 2022 [Undergraduate Mathematics Symposium](#) and Wheaton College’s 2022 Homecoming STEM Poster Session.

Understanding and Modeling Predictors to Earned Run Average in Modern Major League Baseball

- Conducted variable selection using predictors from the Lahman baseball database to model Earned Run Average using regression analysis. Used the same predictors to build a single-layer neural network and a radial support vector regression model in R/RStudio.

EDUCATION

2024 - Pending	PhD Statistical Science Baylor University, Waco, TX	
2023 - 2024	M.S. Statistical Science Baylor University, Waco, TX	(GPA: 3.75/4.0)
2019 - 2023	B.S. Applied Mathematics with Statistics B.A. Economics Wheaton College, Wheaton, IL <i>Edman Presidential Scholarship</i>	(GPA: 3.46/4.0)

SKILLS

Software Tools:	Docker, RStudio, Git/Github, reveal.js, Positron.
Software Languages:	R, JAGS/Stan, Python, C, SAS, Stata, SQL, zsh, L ^A T _E X, JavaScript, CSS, HTML
Relevant Coursework:	Bayesian Methods in Statistics, Survival Analysis, Time Series, Theory of Statistics, Statistical Methods, Computational Statistics, Numerical Analysis, Linear Algebra for Data Analysis, Mathematical Statistics, Machine Learning, Probability Theory, Real Analysis, Intensive Introduction to Computer Science (<i>Harvard</i>), Econometrics, Economics of Labor and Poverty.

EMPLOYMENT

Research Assistant January 2024 - present

Baylor University Department of Statistical Science

- Conducts research with Dr. Mandy Hering, working with outside collaborators in urban water treatment and infrastructure.

Graduate Assistant August 2023 - December 2024

Baylor University Department of Statistical Science

- Collaborated at Baylor University in the Statistical Consulting Center on cutting edge research in various departments.
- Graded and operated as teaching assistant for both the undergraduate and graduate offerings of Computational Statistics (STA 4373/5373/6375) with Dr. David Kahle.
- Lead four supplemental instructorship sections per week for Baylor's Introduction to Statistical Methods course (STA 2381).

Academic Coach January 2023 - Mar 2023

Wheaton Academy Center for Lifelong Learning

- Mentored students 1-on-1 per their request to teach, review, and sharpen their math skills. I met with students ranging in courses from Algebra I to AP Statistics to help them prepare for their exams, whether it's ones for courses or AP exams.

Mathematics Teaching Assistant August 2022 - May 2023

Wheaton College Department of Mathematics and Computer Science

- Served as Applied Machine Learning TA (MATH 465) from December 2022 to present, and the Probability Theory TA (MATH 363) from August 2022 to December 2022.
- Graded weekly homework assignments and hosted weekly help sessions for students.

Participant June 2022 - July 2022

Duke/North Carolina State University Summer Institute for Biostatistics

- Participated in several classroom discussions to engage standard biostatistical methods and areas of study at Duke Clinical Research Institute and North Carolina State University.
- Culminated in a Hackathon studying and modeling myocardial infarction and associated complications. [Link to Github](#) for our Hackathon project.

VOLUNTEER

Intern

June 2023 - August 2023

Křesťanská Akademie Mladých

- Support-raised for four months and built a [blog](#) from scratch to keep supporters in the loop.
- Lived and worked in the Czech Republic for two and a half months while assisting local CB churches by teaching English and sports in schools and at camps. Onboarded and trained short-term mission teams from the U.S. and Ireland who staffed English camps.

HONORS/AWARDS

Varsity Baseball Player

August 2019 - May 2022

Wheaton College Athletics, NCAA

- Practiced baseball six days a week for multiple hours and played over forty-games during the spring semester.

Christian Character Award

May 2019

Valley Christian High School, Chandler AZ

- This award is extended to one graduating Senior at Valley Christian High School who, in their four years at the institution, has most consistently exemplified Christ-like character, lived above reproach, and exhibited a passion for the gospel.